

Causality - SoSe 2025

Toward Structure Learning

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Throughout the lecture, we have touched the following topics:

- Observational Data vs. Randomized Controlled Trials
- Markov Properties and DAG Models (& d-Separation)
- Causal DAG Models and Causal Inference:
 - Parent-Adjustment
 - Path-Method
 - Back-Door and Front-Door Criterion
 - do-Calculus

Definition A path between i and j is **blocked** by a set $S \subseteq V$ (not containing i or j) if

- there is a **non-collider** v on the path, and $v \in S$, or
- there is a **collider** v on the path, and **neither** v **nor any of its descendants** are in S .

A path that is **not blocked** is **active**.

Given two disjoint sets of nodes $A, B \subseteq V$. If **all paths** between **any combination** of nodes $i \in A$ and $j \in B$ are **blocked by S** , then **A and B are d-separated by S** , written as $A \perp\!\!\!\perp_G B \mid S$.

Otherwise, A is **d-connected** to B given S , denoted by $A \not\perp\!\!\!\perp_G B \mid S$.

Note that $\perp\!\!\!\perp_G$ is defined for **graphs** and should not be confused with $\perp\!\!\!\perp$, which relates to **statistical independence**.

So Far

So far, we **relied on**:

- Markov Properties
- d-Separation

to **derive independencies** from **(d-)separations** in the DAG.

In today's lecture, we will learn *the fundamental assumptions for causal discovery*. In particular, we discuss the following concepts:

- Markov Equivalence
- Faithfulness
- Causal Minimality
- Causal Sufficiency
- Markov Blankets
- Grow and Shrink Algorithm

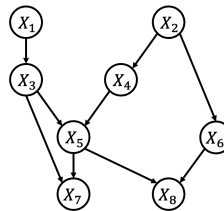
Causal Discovery

Given Observational Data

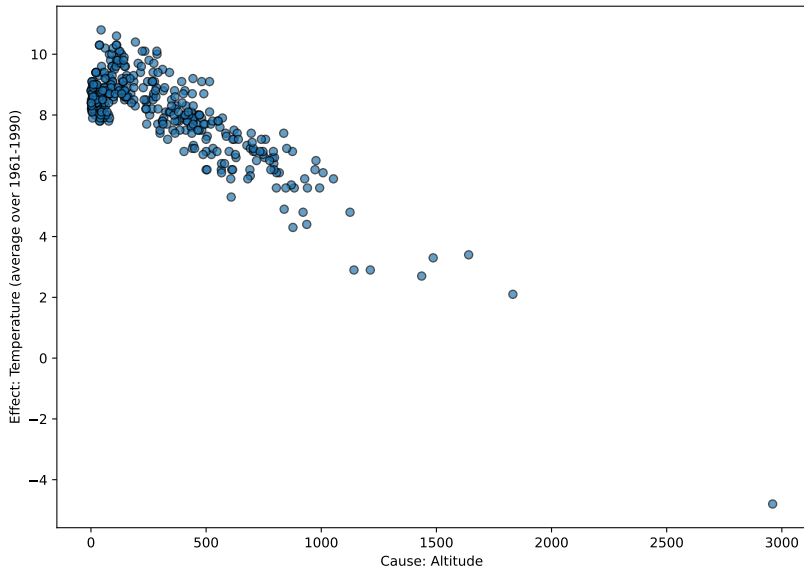
Causal Graph

X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
-1.164294838	0.1315188938	1.58823116792	0.256584649	2.344482611	-0.8508841577	0.187764684	-0.189548880
-1.395184841	0.0713703928	2.0417832952	0.64699484	1.96877622	-0.1808971487	0.7548158889	1.712846466
0.669429152	0.5834298314	-0.2833515157	0.266877236	1.286156029	0.1495888881	1.887956672	-0.415881548
0.281789152	-0.127366437	-1.7297751688	0.1588934615	0.915791977	-0.4805514485	2.814992428	0.255641212
0.85698838	-1.184866493	-0.3318643158	-0.297364895	-0.88482615	0.8603851315	-0.84919182	0.209348383
-0.22823832	-1.223474591	0.2872819233	-0.431887748	-0.3277688	-0.2215134888	-0.854887281	-0.352958855
0.466178978	0.5564627577	0.5897212579	0.571882532	1.43714958	0.8241722327	1.889286148	0.498476879
0.222656156	-0.1236837857	1.695395888	1.705805783	1.68515565	-0.897873183	-0.954513128	-0.602615515
0.386681164	-1.1514173625	2.8688842223	2.256696479	1.1577515	-0.8611917611	-0.8439258815	-0.809388462
0.483298786	1.1984513218	1.6882312814	2.861521916	0.452819161	-0.8878714857	0.778899022	-0.817972379
0.410897488	1.7882178891	0.8767878056	0.870097812	1.67461615	0.8996882879	0.886114281	0.337433051
1.475634379	0.7215880886	2.54614235617	4.815127748	0.153655223	-0.8407818866	0.212861271	-0.255484861
0.588127128	-0.284880188	0.7917425279	-1.251048416	-0.74870984	1.8564616347	-0.258789884	0.738176448
0.76288283	0.8737950882	-0.51598128581	0.234798748	2.83946678	-0.8886848471	-0.888471693	1.168187128
0.542789778	0.8782111134	0.8738748328	0.617638884	1.77887795	0.1756621278	1.398171718	2.299785886
0.077309151	0.8882830833	0.18848168156	0.131322887	-0.7474658	-0.261482195	-0.832495227	-0.147215867
1.18151746	0.1886671111	1.645164888	1.559216753	1.835880003	-0.115653490	-1.84123821	-0.808146671
1.366784215	-0.4449776574	1.5883332127	-0.441391457	1.58496116	0.8146897487	-0.944848494	-0.133623852
0.37177871	-1.3588868714	0.5613788964	0.913148886	1.148184892	0.8238214647	-0.434512089	0.981793964
0.17814711	0.8775880193	-0.6733812865	-0.860548418	1.3352566	-0.8584337577	-1.277670803	0.767949178
-0.41642877	-1.5282162177	-0.9194423637	-1.355881541	1.87161409	-0.8213835888	-0.851284116	0.820824956
-1.480463148	-1.3188156388	-0.6148816365	-0.895691884	1.788484578	-0.8882971542	-0.83185415	-0.861964887
-1.386234819	-0.8327643918	0.2844181933	-0.521884635	0.48523332	0.8979884978	-1.821860285	-0.839874794
-0.216778779	-0.212456178	0.7728161508	0.138454242	1.39815672	-0.197977888	1.185721429	0.287264556
1.644721146	-0.2888852617	-0.8852812833	1.5888021256	1.78277812	-0.8431614688	-0.357492727	0.802884656
-0.436576472	-0.217837882	0.9772814898	0.943677627	-0.11288663	-0.1978812972	0.882491828	-0.308845522
-0.187488936	-0.3877861818	-0.7338742567	0.826485782	-0.88488504	0.8871343846	-0.881748864	-1.173778441
0.818289232	-0.2139548832	0.1588885561	1.817038478	0.85578258	0.8188281487	1.885174728	-1.272232423
-0.360573133	-0.8438878955	0.5128774812	-0.167756178	-0.52181628	-0.6801118871	-1.431762331	1.802524166
-0.494789286	-0.1381758298	-0.1688418857	-0.868356448	2.93733087	-0.8628748757	0.519616173	1.3839382363

Learn



Tübingen Cause-Effect Pair Challenge



Why is Causal Discovery a Hard Problem?

Assume we have a **magic score** S which assigns a positive number to a DAG $\mathcal{G}$. S is **maximal and unique** for the **true DAG** $\mathcal{G}^*$.

Question: If we randomly enumerate all DAGs with d nodes and score them, how many do we need to score?

Intuition:

- $\binom{d}{2} = \frac{d(d-1)}{2}$ distinct vertex combinations
- For each pair (ignoring cycles) there are **three options**: X, Y (no edge), $X \rightarrow Y, X \leftarrow Y$
- For $d = 3$, we get **3** $\binom{d}{2} = 3^3 = 27$
- There can be 2 directed cycles (A, B, C, A) or (A, C, B, A)
- If we subtract them, we get $\#DAGs = 25$ for $d = 3$

DAGs with d Nodes

One can show that the number of DAGs grows **super-exponential** in d !

#Nodes	#Possible DAGs
0	1
1	1
2	3
3	25
4	543
5	29281
6	3781503
7	1138779265
8	783702329343
9	1213442454842881
10	4175098976430598143
11	31603459396418917607425
12	521939651343829405020504063
13	18676600744432035186664816926721
14	1439428141044398334941790719839535103

Figure 1: Figure from: <https://oeis.org/A003024> (February 2022)

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11	31603459396418917607425
12	521939651343829405020504063
13	18676600744432035186664816926721
14	1439428141044398334941790719839535103

Figure 2: Figure from: <https://oeis.org/A003024> (February 2022)

More Questions

Question: How do we score a DAG?

We will postpone this question for now, however...

Question: When are two DAGs similar?

Some notation:

- A **v-structure**, **immortality**, or **unshielded collider** in a DAG $\mathcal{G}$ is a **collider** structure $X \rightarrow Z \leftarrow Y$, where X, Y are **not adjacent** in $\mathcal{G}$.
- The **skeleton** of a DAG $\mathcal{G}$ is an **undirected graph** in which all adjacent nodes in $\mathcal{G}$ are adjacent. *Simply put:* We forget about the edge directions.

Recall “Markov”

Definition Given a DAG $\mathcal{G} = (\mathbf{V}, \mathcal{E})$. A distribution P over $\mathbf{X}_{\mathbf{V}}$ with density p satisfies the following Markov properties:

- **Global Markov property** wrt $\mathcal{G}$ if for all disjoint $\mathbf{A}, \mathbf{B}, \mathbf{C} \subseteq \mathbf{V}$:

$$\mathbf{A} \perp\!\!\!\perp_{\mathcal{G}} \mathbf{B} \mid \mathbf{C} \implies \mathbf{X}_{\mathbf{A}} \perp\!\!\!\perp \mathbf{X}_{\mathbf{B}} \mid \mathbf{X}_{\mathbf{C}}$$

- **Local Markov property** wrt $\mathcal{G}$ if $\forall \mathbf{S} \subseteq \mathbf{V}$

$$\mathbf{X}_{\mathbf{S}} \perp\!\!\!\perp \mathbf{X}_{\text{Nd}(\mathbf{S}) \setminus \text{Pa}(\mathbf{S})} \mid \mathbf{X}_{\text{Pa}(\mathbf{S})}$$

- **Markov factorization property** wrt $\mathcal{G}$ if

$$p(x_1, \dots, x_d) = \prod_{j=1}^d p(x_j \mid x_{\text{Pa}(j)})$$

If P has a density p all three properties are equivalent and we simply say that P is **Markov** to $\mathcal{G}$

Def. Markov Equivalence (ECI 8.12)

1. The set of distributions that are **Markov** wrt a graph $\mathcal{G}$ is

$$\mathcal{M}(\mathcal{G}) := \{P : P \text{ is Markov wrt } \mathcal{G}\}.$$

2. DAGs $\mathcal{G}_1$ and $\mathcal{G}_2$ are **Markov equivalent** if $\mathcal{M}(\mathcal{G}_1) = \mathcal{M}(\mathcal{G}_2)$.
3. The **set of all graphs** that are **Markov equivalent** with some DAG is called the **Markov equivalence class**.

Theorem (ECI 8.13) Two DAGs $\mathcal{G}_1$ and $\mathcal{G}_2$ are **Markov equivalent** if and only if they have the **same skeleton** and the **same immortunities**.

Definition A Markov equivalence class (MEC) of DAGs can be **uniquely represented** by a **completed partially directed acyclic graph (CPDAG)**:

- $i \rightarrow j$ is in the CPDAG iff $i \rightarrow j$ is in all DAGs of the MEC (direct causal effect)
- $i - j$ is in the CPDAG iff there is a DAG in the MEC with $i \rightarrow j$ and one with $i \leftarrow j$ (unidentifiable orientations).

Examples

Which of the following graphs are Markov equivalent?

- a) $X \rightarrow Z \rightarrow Y$
- b) $X \leftarrow Z \leftarrow Y$
- c) $X \leftarrow Z \rightarrow Y$
- d) $X \rightarrow Z \leftarrow Y$

Answer: All graphs except d).

- The CPDAG for graphs a) to c) is $X - Z - Y$
- The CPDAG for graph d) is $X \rightarrow Z \leftarrow Y$

Examples

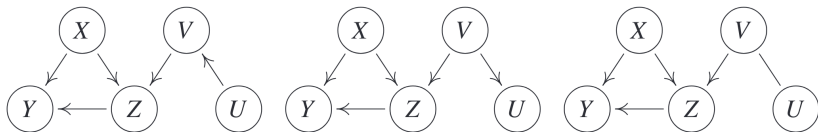


Figure 3: Figure 6.4 (ECI): The graphs on the left and center are Markov equivalent and their CPDAG is on the right.

- Left and center are Markov equivalent
- Right is the corresponding CPDAG

Question: Why are the left and center graphs the **only** ones in the Markov equivalence class?

- $X \rightarrow Z \leftarrow V$ is an immortality (hence fix)
- $Z \rightarrow Y$ is fix since otherwise $Y \rightarrow Z \leftarrow V$ would form an immortality
- Consequently, $X \rightarrow Y$ is the only option that does not induce a cycle
- For (V, U) there are two options (hence undirected in the CPDAG)

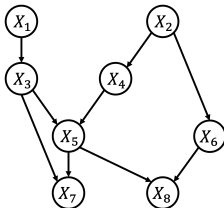
Faithfulness

Back to Causal Discovery

What we know so far:

$$\begin{aligned} X_1 &\perp\!\!\!\perp X_2 \\ X_1 &\perp\!\!\!\perp X_5 \mid X_3 \\ X_3 &\not\perp\!\!\!\perp X_4 \mid X_5 \quad \dots \\ X_3 &\perp\!\!\!\perp X_2 \end{aligned}$$

Markov



I-Map

- The Markov property helps us to connect data (distributions) and graphs by *relating* independencies and d-separations
- By definition, in a **Bayesian network**, P is Markov wrt $\mathcal{G}$
 - Equivalently, we say that $\mathcal{G}$ is an **independence map (I-map)** of P
- Every distribution is Markov wrt a fully connected DAG
 - Equivalently, a fully connected DAG is an **I-map** of any distribution over the respective variables

Back to Causal Discovery

What we want:

Given Observational Data

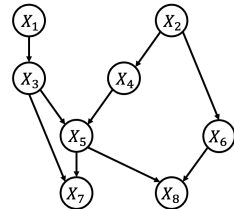
Causal Graph

$X_1 \quad X_2 \quad X_3 \quad X_4 \quad X_5 \quad X_6 \quad X_7 \quad X_8$

```

-1.144394830 0.1315338938 1.3882314792 0.236836649 2.36448261 -0.8508842672 0.187764094 -0.189348888
-1.395158451 0.873232926 2.0417212902 0.646594844 1.94577622 -0.1809292483 0.754815089 1.212945466
0.660429122 0.4834309314 -0.2833131917 0.386077216 1.296189269 0.1605888861 -1.487996472 -0.4198811648
0.281788913 -0.4327864367 -1.7327913688 -1.538981615 0.9191917 -0.8506414485 0.211693028 0.8164413132
0.850498918 -1.1242846952 -0.2518643219 -0.1977638893 -0.8845219 -0.0804851215 -0.84913282 0.298348263
-0.228828822 -1.2234724991 -0.2872639233 -0.431887745 -0.32275688 -0.2215118486 -0.854887283 -0.3329198355
0.444178876 0.5844407567 0.6087121579 1.971888212 1.43744608 0.8041723227 -1.498204448 0.498496898
0.222064134 -0.1263857857 1.4963989809 1.718895783 1.44932649 -0.899728183 -0.93461338 -0.8508365315
0.180584246 -1.1144373505 2.088912333 2.256494479 3.11767154 -0.6411971631 -0.842924815 -0.649388462
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0.598722226 -0.3988582186 0.7291182779 1.510498456 -0.74879284 0.8564442437 -0.258799284 0.7336174238
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1.356738215 -0.4449775274 1.1861311223 1.284197427 1.58416174 0.816589489 -0.94384829 -0.133613322
0.371778971 -1.3586686714 0.613748062 -0.913148844 1.54874692 0.653811447 -0.436135899 0.90174964
-0.175147311 0.8719881983 -0.6939313865 -0.868544518 1.39353564 0.830437377 1.277688952 0.967649378
-0.41628877 -1.5282153177 -0.9794620237 -1.320881541 1.87161697 -0.821381886 -0.821261114 0.820829556
-1.582651613 -1.3218626288 0.6764882365 0.497958286 3.78848078 -0.6862921262 -0.811856415 -0.862861887
-1.196231629 -0.821764918 0.2864381955 -0.521884635 0.43523552 0.497088449 -1.812464065 -0.808717894
-0.11677879 0.7421158878 -0.6574313638 -0.876264862 -1.33815587 -0.191884497 -0.59979088 -1.587931429
1.644721146 -0.288806787 0.288806787 1.588806787 1.78571888 -0.8421656612 0.25747727 -0.50285656
-0.836576472 -0.2378167982 0.9772548908 0.942671627 -0.11289623 -0.199984272 0.482491828 -0.368455262
-0.18188586 -0.188186148 0.738714197 -0.86486791 -0.8848804 0.897134366 -0.88174864 -1.17378641
0.61829922 -0.251954032 0.158888551 1.817038478 0.85578256 0.8188291487 1.885174728 -1.273252433
-0.34573233 -0.8683878955 0.5118271882 0.16725478 0.21224208 -0.680311871 -1.437473333 1.802914166
-0.491789216 -1.138178878 0.5484717857 0.862356146 2.92712387 -0.862071877 0.51961473 1.3883982363
0.104192842 0.184787891 0.1497889774 0.817614174 1.77571623 0.8507481528 -0.84224187 -1.765357179
    
```

Learn



"If I find an independence, there is no edge in $\mathcal{G}$ "

$$X_A \perp\!\!\!\perp X_B \mid X_C \implies A \perp\!\!\!\perp_{\mathcal{G}} B \mid C$$

Faithfulness and Causal Minimality

Definition Given a DAG $\mathcal{G} = (\mathbf{V}, \mathcal{E})$, and a distribution P over $\mathbf{X}$ with density p .

1. P is said to be **faithful** to $\mathcal{G}$ if for all pairwise disjoint subsets $\mathbf{A}, \mathbf{B}, \mathbf{C} \subseteq \mathbf{V}$:

$$\mathbf{X}_\mathbf{A} \perp\!\!\!\perp \mathbf{X}_\mathbf{B} \mid \mathbf{X}_\mathbf{C} \implies \mathbf{A} \perp\!\!\!\perp_{\mathcal{G}} \mathbf{B} \mid \mathbf{C} .$$

2. P satisfies **causal minimality** wrt $\mathcal{G}$ if it is Markovian with respect to $\mathcal{G}$, but not to any proper subgraph $\mathcal{G}' = (\mathbf{V}, \mathcal{E}')$ of $\mathcal{G}$ with $\mathcal{E}' \subset \mathcal{E}$.

Alternatively, if $\mathcal{G}$ is an **I-map** to P but no proper subgraph $\mathcal{G}'$ of $\mathcal{G}$ is an I-map to P , then P **satisfies causal minimality** wrt $\mathcal{G}$.

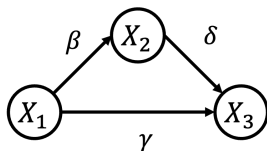
Faithfulness (Example)

Consider a linear Gaussian SCM:

$$X_1 := N_1$$

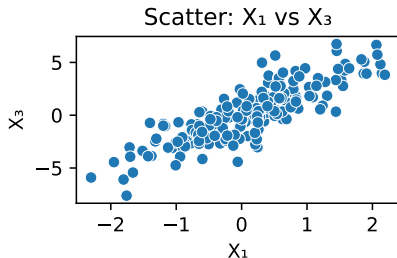
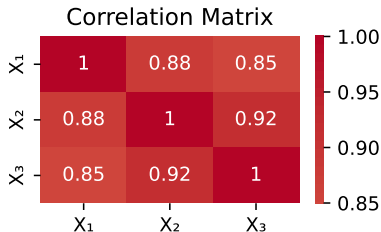
$$X_2 := 2X_1 + N_2$$

$$X_3 := 0.5X_1 + X_2 + N_3$$



All $N_i \sim \mathcal{N}(0, 1)$, pairwise independent.

Clearly, for all pairs, **faithfulness holds** in this example.



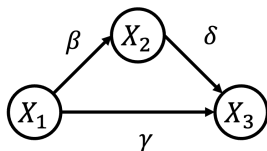
Faithfulness (Cancelling Path)

Set $\gamma = -(\beta\delta)$ in the SCM:

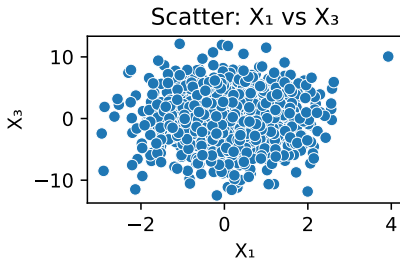
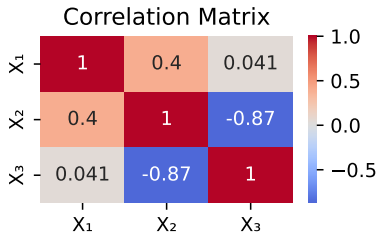
$$X_1 := N_1$$

$$X_2 := 0.5X_1 + N_2$$

$$X_3 := 2X_1 - 4X_2 + N_3$$



All $N_i \sim \mathcal{N}(0, 1)$, pairwise independent.



Faithfulness (Cancelling Path)

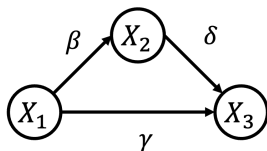
Set $\gamma = -(\beta\delta)$ in the SCM:

$$X_1 := N_1$$

$$X_2 := 0.5X_1 + N_2$$

$$X_3 := 2X_1 - 4X_2 + N_3$$

$$= \underbrace{2X_1 - 4(0.5X_1 + N_2)}_{=0X_1 - 4N_2} + N_3$$



Consequently, on population level, we get that

$$X_1 \perp\!\!\!\perp X_3 \text{ while } 1 \not\perp_{\mathcal{G}} 3 \quad \textcolor{red}{\downarrow}$$

i.e., P is **not** faithful wrt $\mathcal{G}$!

The example above is known as **cancelling path** and is a well-known violation of the faithfulness assumption.

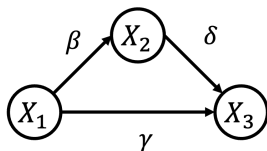
Causal Minimality (Cancelling Path)

Set $\gamma = -(\beta\delta)$ in the SCM:

$$X_1 := N_1$$

$$X_2 := 0.5X_1 + N_2$$

$$X_3 := 2X_1 - 4X_2 + N_3$$



Question: Does P satisfy causal minimality?

- Let $\mathcal{G}_2$ be the v-structure $X_1 \rightarrow X_2 \leftarrow X_3$
- P is Markov to $\mathcal{G}_2$: the only d-separation in $\mathcal{G}_2$, i.e., $X_1 \perp\!\!\!\perp_{\mathcal{G}_2} X_3$ is reflected by P
- $\mathcal{G}_2$ is not a proper subgraph of $\mathcal{G}$ (all good so far)
- However, P is also Markov to $\mathcal{G}_3$, which is $\mathcal{G}_2$ plus the edge $X_1 \rightarrow X_3$: in $\mathcal{G}_3$ there are no d-separations
- Clearly, $\mathcal{G}_2 \subset \mathcal{G}_3$, and hence causal minimality does not hold.

Faithfulness implies Causal Minimality

Proposition (Faithfulness $\Rightarrow$ Minimality, ECI book, 6.35) If P is faithful and Markov wrt $\mathcal{G}$, then causal minimality is satisfied.

Proof Sketch:

- Construct $\tilde{\mathcal{G}} \subset \mathcal{G}$ by deleting an edge from $\mathcal{G}$ and assume P is Markov wrt $\tilde{\mathcal{G}}$ (violation of minimality)
- Hence, $\tilde{\mathcal{G}}$ implies an additional d-separation for a pair of nodes (i, j) ; by the Markov condition, an additional independence
- Thus, P cannot be faithful wrt $\mathcal{G}$

Faithfulness implies Causal Minimality

Proposition (Faithfulness $\Rightarrow$ Minimality, ECI book, 6.35) If P is faithful and Markov wrt $\mathcal{G}$, then causal minimality is satisfied.

Perhaps a simpler variant of causal minimality is given by stating that **no parent** of a node k **should be independent** of k **when we condition on all remaining parents**:

Proposition (ECI book, 6.36) Suppose $(\mathcal{G}, P)$ is a DAG model (P is Markov wrt to $\mathcal{G}$). Then, P satisfies **causal minimality** with respect to $\mathcal{G}$ if and only if $\forall k \in \mathbf{V} : \forall j \in \text{Pa}(k) :$

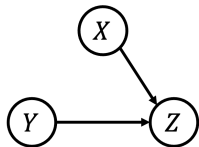
$$X_k \not\perp\!\!\!\perp X_j \mid X_{\text{Pa}(k) \setminus \{j\}} .$$

Next, we consider an example that violates faithfulness but satisfies causal minimality.

Two Coins and a Bell

Setup:

- Let X, Y be two independent coins with probability of heads equal to p and q resp.
- Z is a bell which rings if **only** X **or** Y shows heads, i.e. $Z := X \oplus Y$



Let's investigate P , especially for the case where both are fair coins:

X	Y	Z	$P \ (p=q=1/2)$
0	0	0	$(1-p)(1-q) = 1/4$
0	1	1	$(1-p)q = 1/4$
1	0	1	$p(1-q) = 1/4$
1	1	0	$pq = 1/4$

$$\begin{aligned} p(X=1) &= p(X=1|Z=1) \\ &= p(X=1|Z=0) = 1/2 \end{aligned}$$

Thus, $X \perp\!\!\!\perp Z$, the same holds for Y and Z , and $X \perp\!\!\!\perp Y$ by assumption.

We can conclude that **faithfulness** is clearly **violated** since $X \not\perp\!\!\!\perp_{\mathcal{G}} Z$, etc. Note, that this example **does not violate** faithfulness for $p, q \neq 1/2$.

Two Coins and a Bell

What about **causal minimality**? Let's condition on the remaining parents:

X	Y	Z	$P (p=q=1/2)$
0	0	0	$(1-p)(1-q) = 1/4$
0	1	1	$(1-p)q = 1/4$
1	0	1	$p(1-q) = 1/4$
1	1	0	$pq = 1/4$

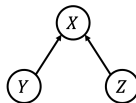
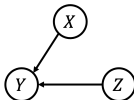
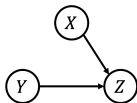
$$p(X = 1|Y = 1) = 1/2$$

$$p(X = 1|Y = 1, Z = 1) = 0$$

$$p(Y = 1|X = 1, Z = 1) = 0$$

$$p(Z = 1|X = 1, Y = 1) = 0$$

- We **cannot find any conditional independence**, i.e., $X \not\perp\!\!\!\perp Y | Z$, $X \not\perp\!\!\!\perp Z | Y$, and $Y \not\perp\!\!\!\perp Z | X$
- P is Markov to any of the DAGs below, and to no proper sub-graph of any of them $\implies$ **causal minimality holds**



Remarks to Faithfulness Violations

As shown previously, **faithfulness** should not be assumed to always hold, **however, we tried very hard** to find violations:

- Cancelling path: $\gamma = -(\beta\delta)$ **needs to hold exactly**
- XOR: p or q **needs to be exactly** $1/2$
 - Note that also noisy XORs (SCMs) can be constructed (see Marx, Gretton, Mooij, 2021)
- In general, faithfulness violations **have measure zero** in linear systems (Meek, 1995)

In practice, one still needs to be careful since testing conditional independencies from finite data is hard, too (Shah and Peters, 2020).

Toward Discovery: Markov Blankets

Causal Sufficiency

Since causal discovery, when observing all relevant nodes, is already difficult, we exclude more difficult cases for now and **assume causal sufficiency** if not stated otherwise:

Definition (Spirtes et al. 2000) A set of variables $\mathbf{V}$ is said to be **causally sufficient** if there is **no hidden common cause** $c \notin \mathbf{V}$ that is causing more than one variable in $\mathbf{V}$, i.e., there exists no unobserved confounder.

Markov Blanket

Let's start small and consider local structures. . .

Question: Given node A in a DAG, what subset of nodes *isolates* A from the rest of the network?

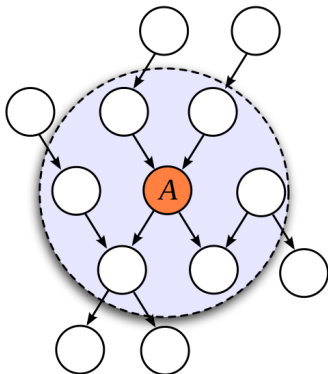


Figure 4: Source: Wikipedia (https://en.wikipedia.org/wiki/Markov_blanket)

Definition (EC1, 6.26) Given a DAG $\mathcal{G}$ and a target node $k \in V$:

1. The **Markov blanket** $\text{MB}(k)$ of k is the **smallest set** so that

$$k \perp\!\!\!\perp_{\mathcal{G}} V \setminus (\{k\} \cup \text{MB}(k)) \mid \text{MB}(k)$$

i.e., it d-separates all other nodes from k .

2. If P is Markov wrt $\mathcal{G}$, then

$$X_k \perp\!\!\!\perp X_{V \setminus (\{k\} \cup \text{MB}(k))} \mid X_{\text{MB}(k)} .$$

Markov Blanket

One can show (ECI, 6.27) that the **Markov blanket** of a node k consists of its **parents**, **children**, and **spouses** (nodes that have a common child node with k):

$$\text{MB}(k) = \text{Pa}(k) \cup \text{Ch}(k) \cup \text{Sp}(k) .$$

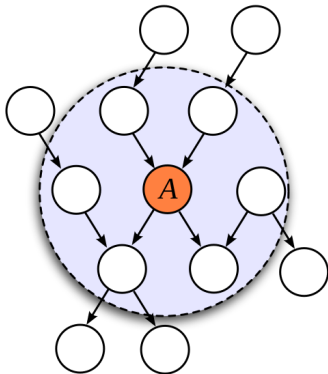


Figure 5: Source: Wikipedia (https://en.wikipedia.org/wiki/Markov_blanket)

Putting everything together: Learning the Markov Blanket

Grow and Shrink Algorithm¹

Algorithm 1: GS for Markov Blankets

input : $\mathbf{X}_V$ with joint P , Target $t \in V$

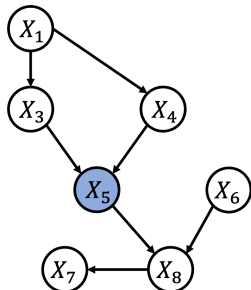
output : $\text{MB}(t)$

```
1  $\mathbf{S} \leftarrow \emptyset$ ;  
  // Grow Phase  
2 while  $(\exists k \in V \setminus \{t\} : X_t \not\perp\!\!\!\perp X_k \mid \mathbf{X}_S)$  do  
3    $\mathbf{S} \leftarrow \mathbf{S} \cup \{k\}$  ;  
  // Shrink Phase  
4 while  $\exists k \in \mathbf{S} : X_t \perp\!\!\!\perp X_k \mid \mathbf{X}_{S \setminus \{k\}}$  do  
5    $\mathbf{S} \leftarrow \mathbf{S} \setminus \{k\}$  ;  
6 return  $\mathbf{S}$ 
```

Example Run ($t = 5$):

Initialization

$\mathbf{S} = \emptyset$



P is faithful and Markov wrt $\mathcal{G}$.

¹Margaritis, Thrun (1999)

Grow and Shrink Algorithm

Algorithm 1: GS for Markov Blankets

input : $\mathbf{X}_V$ with joint P , Target $t \in V$

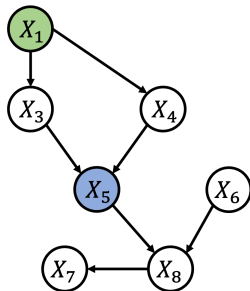
output : $\text{MB}(t)$

```
1  $\mathbf{S} \leftarrow \emptyset$ ;  
  // Grow Phase  
2 while  $(\exists k \in V \setminus \{t\} : X_t \not\perp\!\!\!\perp X_k \mid \mathbf{X}_S)$  do  
3    $\mathbf{S} \leftarrow \mathbf{S} \cup \{k\}$  ;  
  // Shrink Phase  
4 while  $\exists k \in \mathbf{S} : X_t \perp\!\!\!\perp X_k \mid \mathbf{X}_{S \setminus \{k\}}$  do  
5    $\mathbf{S} \leftarrow \mathbf{S} \setminus \{k\}$  ;  
6 return  $\mathbf{S}$ 
```

Example Run ($t = 5$):

GP: $X_5 \not\perp\!\!\!\perp X_1$

$\mathbf{S} = \{1\}$



P is faithful and Markov wrt $\mathcal{G}$.

Grow and Shrink Algorithm

Algorithm 1: GS for Markov Blankets

input : $\mathbf{X}_V$ with joint P , Target $t \in V$

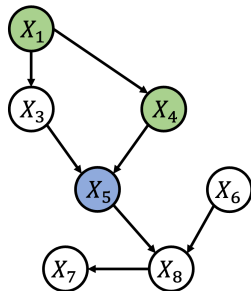
output : $\text{MB}(t)$

```
1  $\mathbf{S} \leftarrow \emptyset$ ;  
  // Grow Phase  
2 while  $(\exists k \in V \setminus \{t\} : X_t \not\perp\!\!\!\perp X_k \mid \mathbf{X}_S)$  do  
3    $\mathbf{S} \leftarrow \mathbf{S} \cup \{k\}$  ;  
  // Shrink Phase  
4 while  $\exists k \in \mathbf{S} : X_t \perp\!\!\!\perp X_k \mid \mathbf{X}_{S \setminus \{k\}}$  do  
5    $\mathbf{S} \leftarrow \mathbf{S} \setminus \{k\}$  ;  
6 return  $\mathbf{S}$ 
```

Example Run ($t = 5$):

GP: $X_5 \not\perp\!\!\!\perp X_4 \mid \{X_1\}$

$\mathbf{S} = \{1, 4\}$



P is faithful and Markov wrt $\mathcal{G}$.

Grow and Shrink Algorithm

Algorithm 1: GS for Markov Blankets

input : $\mathbf{X}_V$ with joint P , Target $t \in V$

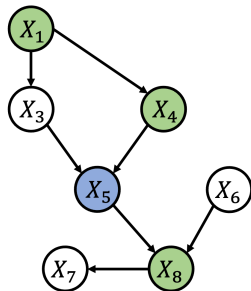
output : $\text{MB}(t)$

```
1  $\mathbf{S} \leftarrow \emptyset$ ;  
  // Grow Phase  
2 while  $(\exists k \in V \setminus \{t\} : X_t \not\perp\!\!\!\perp X_k \mid \mathbf{X}_S)$  do  
3    $\mathbf{S} \leftarrow \mathbf{S} \cup \{k\}$  ;  
  // Shrink Phase  
4 while  $\exists k \in \mathbf{S} : X_t \perp\!\!\!\perp X_k \mid \mathbf{X}_{S \setminus \{k\}}$  do  
5    $\mathbf{S} \leftarrow \mathbf{S} \setminus \{k\}$  ;  
6 return  $\mathbf{S}$ 
```

Example Run ($t = 5$):

GP: $X_5 \not\perp\!\!\!\perp X_8 \mid \{X_1, X_4\}$

$\mathbf{S} = \{1, 4, 8\}$



P is faithful and Markov wrt $\mathcal{G}$.

Grow and Shrink Algorithm

Algorithm 1: GS for Markov Blankets

input : $\mathbf{X}_V$ with joint P , Target $t \in V$

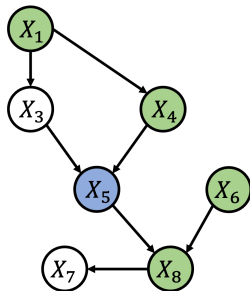
output : $\text{MB}(t)$

```
1  $\mathbf{S} \leftarrow \emptyset$ ;  
  // Grow Phase  
2 while  $(\exists k \in V \setminus \{t\} : X_t \not\perp\!\!\!\perp X_k \mid \mathbf{X}_S)$  do  
3    $\mathbf{S} \leftarrow \mathbf{S} \cup \{k\}$  ;  
  // Shrink Phase  
4 while  $\exists k \in \mathbf{S} : X_t \perp\!\!\!\perp X_k \mid \mathbf{X}_{S \setminus \{k\}}$  do  
5    $\mathbf{S} \leftarrow \mathbf{S} \setminus \{k\}$  ;  
6 return  $\mathbf{S}$ 
```

Example Run ($t = 5$):

GP: $X_5 \not\perp\!\!\!\perp X_6 \mid \{X_1, X_4, X_8\}$

$\mathbf{S} = \{1, 4, 6, 8\}$



P is faithful and Markov wrt $\mathcal{G}$.

Grow and Shrink Algorithm

Algorithm 1: GS for Markov Blankets

input : $\mathbf{X}_V$ with joint P , Target $t \in V$

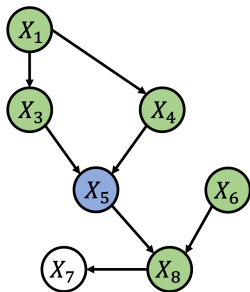
output : $\text{MB}(t)$

```
1  $\mathbf{S} \leftarrow \emptyset$ ;  
  // Grow Phase  
2 while  $(\exists k \in V \setminus \{t\} : X_t \not\perp\!\!\!\perp X_k \mid \mathbf{X}_S)$  do  
3    $\mathbf{S} \leftarrow \mathbf{S} \cup \{k\}$  ;  
  // Shrink Phase  
4 while  $\exists k \in \mathbf{S} : X_t \perp\!\!\!\perp X_k \mid \mathbf{X}_{S \setminus \{k\}}$  do  
5    $\mathbf{S} \leftarrow \mathbf{S} \setminus \{k\}$  ;  
6 return  $\mathbf{S}$ 
```

Example Run ($t = 5$):

GP: $X_5 \not\perp\!\!\!\perp X_3 \mid \{X_1, X_4, X_6, X_8\}$

$\mathbf{S} = \{1, 3, 4, 6, 8\}$



P is faithful and Markov wrt $\mathcal{G}$.

Grow and Shrink Algorithm

Algorithm 1: GS for Markov Blankets

input : $\mathbf{X}_V$ with joint P , Target $t \in V$

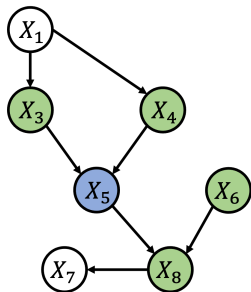
output : $\text{MB}(t)$

```
1  $\mathbf{S} \leftarrow \emptyset$ ;  
  // Grow Phase  
2 while  $(\exists k \in V \setminus \{t\} : X_t \not\perp\!\!\!\perp X_k \mid \mathbf{X}_S)$  do  
3    $\mathbf{S} \leftarrow \mathbf{S} \cup \{k\}$  ;  
  // Shrink Phase  
4 while  $\exists k \in \mathbf{S} : X_t \perp\!\!\!\perp X_k \mid \mathbf{X}_{S \setminus \{k\}}$  do  
5    $\mathbf{S} \leftarrow \mathbf{S} \setminus \{k\}$  ;  
6 return  $\mathbf{S}$ 
```

Example Run ($t = 5$):

SP: $X_5 \perp\!\!\!\perp X_1 \mid \{X_3, X_4, X_6, X_8\}$

$\mathbf{S} = \{3, 4, 6, 8\} = \text{MB}(5)$



P is faithful and Markov wrt $\mathcal{G}$.

Theorem Let $(\mathcal{G}, P)$ be a **DAG model**, where P is **faithful** wrt $\mathcal{G}$. Further, assume access to a **perfect (conditional) independence test**. For any target node $t \in \mathbf{V}$, the Grow and Shrink algorithm (Algorithm 1) returns the **correct** Markov blanket $\text{MB}(t)$.

Notes:

- The GS algorithm can be extended to handle some **faithfulness violations**, e.g., due to XOR (Marx, Gretton, Mooij, 2021)
- The GS algorithm is **not** very **runtime efficient**; faster algorithms are discussed in Penã et al. (2007)

- **Recommended reading** is *ECI* Chapter 6.5 to 6.8
- This course *builds upon and extends* the material from “*Causality*” (WiSe 2022/23) that was kindly provided by **Stefan Harmeling**
- This lecture, *builds upon and extends* the slides of the course “*Causality*” (2021) by Christina Heinze-Deml

Additional References

- Margaritis, Thrun. *Bayesian Network Induction via Local Neighborhoods*. NeurIPS, 1999
- Marx, Mooij, Gretton. *A weaker faithfulness assumption based on triple interactions*. UAI, 2021.
- Meek. *Strong completeness and faithfulness in Bayesian networks*. UAI, 1995
- Penã et al. *Towards scalable and data efficient learning of Markov boundaries*. IJAR, 2007
- Shah, Peters. *The hardness of conditional independence testing and the generalised covariance measure*. Ann. Stat, 2020

Thank you!